

MetrAI — Product Requirements Document

1. Document Purpose

- This Product Requirements Document defines the requirements for MetrAI, a software system for checking compliance of packaged commodities under the Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products, images, labels and product information.
- The document defines the product scope, inspection workflow, compliance checks, reports, repository, dashboard and user access requirements described in the supplied project material and problem statement.

2. Background

- Packaged commodities are widely sold through retail stores, supermarkets and e-commerce platforms across India. Under the Legal Metrology Act, 2009 and the Legal Metrology (Packaged Commodities) Rules, 2011, packaged commodities are required to bear mandatory declarations such as the name and address of manufacturer/packer/importer, net quantity, Maximum Retail Price (MRP), month and year of manufacture/packing/import, consumer care details and other prescribed declarations in the specified format and manner.
- Manual inspection and compliance checking can be time-consuming and resource intensive because of the large volume and variety of packaged products available in the market. Non-compliance such as missing declarations, incorrect font sizes, improper MRP declarations and other non-compliant practices can occur.
- MetrAI is intended to provide a software system capable of scanning product labels, package images and product listings to identify potential violations through image and label analysis.

3. Problem Statement

- Develop a software application capable of scanning packaged commodity labels, product images and product information to automatically assess compliance with the Legal Metrology (Packaged Commodities) Rules, 2011.
- The system shall automatically detect, extract and validate mandatory declarations and identify non-compliances in packaged commodities through image and label analysis.

4. Product Vision

- MetrAI provides a digital compliance-checking workflow for packaged commodities by scanning product images, labels and product information, extracting mandatory declarations, validating them against the applicable Legal Metrology (Packaged Commodities) Rules, 2011, and presenting compliance results, supporting evidence, reports and inspection history.

5. Objectives

- Scan and analyze images of packaged commodities.
- Scan and analyze product labels and product information.
- Detect mandatory declarations prescribed under the Legal Metrology (Packaged Commodities) Rules, 2011.
- Check correctness, completeness and placement of declarations.
- Identify missing or non-compliant declarations.
- Check readability and font size requirements.
- Identify missing, misleading or non-standard declarations.
- Generate compliance reports and violation summaries.
- Maintain a repository of scanned products and compliance history.
- Provide dashboards for enforcement officials.
- Provide search and retrieval for previously scanned products and reports.

6. Scope

- Packaged commodity product scanning.
- Product image and package label analysis.
- Product information and product listing analysis.
- Automated extraction and validation of mandatory declarations.
- Rule-based compliance checking under the Legal Metrology (Packaged Commodities) Rules, 2011.
- Compliance and non-compliance reporting.
- Photographs and supporting evidence.
- Scanned-product repository and inspection history.
- Dashboard monitoring.
- Role-based user access and secure authentication.
- PDF and editable report export.

7. Inspection Inputs

- Images of packaged commodities.
- Images of product labels and packaging.
- Product information.
- Product listings from e-commerce platforms.
- Photographs and supporting evidence.

8. Inspection Workflow

The overall workflow is:

- Physical Product / E-Commerce Listing → Input & Image Quality → AI Evidence Extraction → Structured Product Record → Barcode / Product Cross-Check → Product Classification → Inspection Context → Rule Applicability Engine → Versioned Legal Rule Engine → Evidence Mapping → Explainable Compliance Result → Human Officer Review → Final Decision → Compliance Report / Case Creation / Escalation / Request → Dashboard + Repository → Audit Trail.
- The workflow supports both physical packaged-product inspection and e-commerce/product-listing inspection described in the supplied project material.

9. Image and Label Analysis

- The system shall scan and analyze images of packaged commodities.
- The system shall analyze product labels and package information.
- The system shall support extraction of declarations from labels and packaging.
- The system shall support analysis of product images and product information.
- The system shall support detection of mandatory declarations.

10. Mandatory Declaration Extraction

- The system shall detect and extract mandatory declarations required under the Legal Metrology (Packaged Commodities) Rules, 2011.

The declarations identified in the supplied problem statement include:

- Name and address of manufacturer/packer/importer.

- Net quantity.
- Maximum Retail Price (MRP).
- Month and year of manufacture.
- Month and year of packing.
- Month and year of import, where applicable.
- Consumer care details.
- Other prescribed declarations.

11. Compliance Validation

- The system shall check the correctness of extracted declarations.
- The system shall check the completeness of mandatory declarations.
- The system shall check the placement of declarations.
- The system shall identify missing declarations.
- The system shall identify non-compliant declarations.
- The system shall identify missing, misleading or non-standard declarations.
- The system shall check MRP declarations and identify improper or non-compliant MRP declarations.
- The system shall check readability requirements.
- The system shall check font size requirements.

12. Font Size / Character Height Check

- Detect text bounding boxes.
- Estimate character height.
- Use calibration/reference where available.
- Compare against applicable requirement.
- Borderline → REVIEW REQUIRED.

13. Product Classification and Inspection Context

Product Classification is followed by an explicit Inspection Context stage in the supplied workflow.

- Product type.
- Pack type.
- Sale channel.
- Intended market.
- Inspection date.

14. Rule-Based Compliance Checking

- Compliance checking shall be based on the Legal Metrology (Packaged Commodities) Rules, 2011.
- The system shall use rule-based compliance checking to validate mandatory declarations and identify non-compliances.
- The compliance checking workflow shall use the applicable requirements for the product being inspected.

The workflow includes a Versioned Legal Rule Engine.

The Rule Repository includes the Source / Amendment Ref.

Official source verification is included so that the system is not inventing legal rules.

15. AI / Automated Extraction Workflow

- The supplied project material describes an automated evidence-processing workflow using image analysis, OCR and NLP/NER for extracting information from product evidence.
- The automated layer is intended to detect and extract product declarations and provide information for rule-based compliance checking.

16. Human Review

- The system shall support review of automated extraction and compliance results by the user/enforcement official.
- The automated result shall provide supporting evidence for the compliance decision.

17. Evidence Mapping

Evidence Mapping links rule evaluation to the evidence used for the compliance result.

- Source image.
- OCR text.
- Bounding box.
- Barcode data.
- Detected value.
- Confidence.

18. Compliance Results

- The system shall provide compliance/non-compliance results.
- The system shall provide violation summaries.
- The system shall identify the declarations or requirements associated with a non-compliance result.
- The system shall retain supporting photographs/evidence with the inspection.

19. Compliance Reports

- The system shall generate digital compliance reports.
- The system shall generate non-compliance/violation reports or summaries.
- Reports shall support PDF format.
- Reports shall support an editable format.
- Reports shall include photographs and supporting evidence where applicable.

20. Product Repository and Inspection History

- The system shall maintain a repository of scanned products.
- The system shall maintain compliance and inspection history.
- The system shall provide search and retrieval of previously scanned products.
- The system shall provide search and retrieval of previously generated reports.

21. Dashboard

- The system shall provide dashboards for enforcement officials.
- The dashboard shall support monitoring of inspections, violations and product compliance details.

- The dashboard shall support monitoring of compliance status and enforcement activities.

22. User Access and Authentication

- The system shall provide role-based user access.
- The system shall provide secure authentication.
- Access to application functions shall be controlled according to the user's role.

23. Key Functional Requirements

- FR-01: Image upload and product scanning functionality.
- FR-02: Extraction of declarations from labels and packaging.
- FR-03: Detection of mandatory declarations.
- FR-04: Extraction and analysis of product information and product listings.
- FR-05: Font size analysis.
- FR-06: Readability analysis.
- FR-07: Checking correctness, completeness and placement of declarations.
- FR-08: Detection of missing declarations.
- FR-09: Detection of misleading or non-standard declarations.
- FR-10: Detection of non-compliant declarations.
- FR-11: Checking of MRP declarations, including improper/non-compliant MRP declarations.
- FR-12: Generation of compliance/non-compliance reports.
- FR-13: Attachment of photographs and supporting evidence.
- FR-14: Repository of scanned products and inspection history.
- FR-15: Role-based user access.
- FR-16: Secure authentication.
- FR-17: Dashboard for compliance status, inspections, violations, product compliance details and enforcement activities.
- FR-18: Search and retrieval of previously scanned products and reports.
- FR-19: Export of reports to PDF.
- FR-20: Export of reports to editable formats.

24. Expected Solution

- The proposed solution is a user-friendly web and/or mobile-based software application.
- It shall provide automated extraction and validation of mandatory declarations.
- It shall provide rule-based compliance checking for the Legal Metrology (Packaged Commodities) Rules, 2011.
- It shall generate digital compliance reports in PDF and editable formats.
- It shall provide dashboards for monitoring inspections, violations and product compliance details.
- It shall provide search and retrieval for previously scanned products and reports.
- Technical documentation shall describe the software architecture and deployment framework.

25. Product Outputs

- Extracted mandatory declarations.

- Compliance/non-compliance result.
- Violation summary.
- Photographs and supporting evidence.
- Compliance report.
- Non-compliance report/summary.
- Scanned product record.
- Inspection/compliance history.
- Dashboard information.

26. Technical Documentation Requirement

- The project shall include technical documentation describing the software architecture and deployment framework. The detailed architecture and deployment specifications are maintained separately from this PRD.

27. Scope Boundary

- This PRD contains only the requirements described in the supplied problem statement and MetrAI project material. Technical implementation choices, database design, API contracts, deployment configuration and other implementation details are to be defined in their respective project documents.

28. Requirement Traceability

- The requirements in this PRD directly map to the supplied problem statement's Background, Description, Expected Solution and Key Functional Requirements, together with the supplied MetrAI project workflow and architecture material.

29. Product Requirement Summary

- MetrAI shall provide a software application that scans packaged commodity products, images, labels and product information; automatically extracts and validates mandatory declarations; checks compliance under the Legal Metrology (Packaged Commodities) Rules, 2011; identifies missing, misleading, non-standard or otherwise non-compliant declarations; checks font size, readability and placement; uses Inspection Context and a Versioned Legal Rule Engine; maps evidence including source image, OCR text, bounding box, barcode data, detected value and confidence; supports Human Officer Review and a final inspection decision after officer review; and provides compliance reports, case creation, escalation/request, dashboard + repository, and audit trail.

30. Cross-Cutting Capabilities

- RBAC.
- Security & Privacy.
- Offline Support.
- Multilingual OCR.
- Rule / Amendment Management.
- Audit & Logs.

Document Status

Version: Final

Status: Final selected PRD with supplied workflow refinements

Basis: User-provided MetrAI project documents, workflow/architecture diagrams, and the supplied official problem statement.

Scope rule: No additional product requirements have been introduced beyond the supplied source material.

Field	Details
Product	MetrAI
Problem Statement	Software System to check compliance of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products, images and labels.
Domain	Legal Metrology / Compliance
Document Version	1.1
Status	MVP Product Requirements
Source Basis	Provided MetrAI project documents, workflow/architecture diagrams, and official problem statement supplied by the user